

5.2.2

For each positive integer n , let $P(n)$ be the formula

$$1 + 3 + 5 + \dots + (2n - 1) = n^2$$

- a. Write $P(1)$. Is $P(1)$ true?
- b. Write $P(k)$.
- c. Write $P(k + 1)$.
- d. In a proof by mathematical induction that the formula holds for every integer $n \geq 1$, what must be shown in the inductive step?

Let $P(n)$ be

$$\sum_{i=1}^n 2i - 1 = n^2$$

a.

For the left hand side:

$$P(1) = \sum_{i=1}^1 2i - 1 = (2 \times 1) - 1 = 2 - 1 = \boxed{1}$$

For the right hand side:

$$(1)^2 = 1^2 = \boxed{1}$$

Solution: $P(1)$ is true.

b.

For the left hand side:

$$P(k) = \sum_{i=1}^k 2i - 1 = \boxed{1 + 3 + 5 + \dots + (2k - 1)}$$

For the right hand side: $(k)^2 = \boxed{k^2}$

Solution: $P(k) = 1 + 3 + 5 + \dots + (2k - 1) = k^2$

c.

For the left hand side:

$$P(k+1) = \sum_{i=1}^{k+1} 2i - 1 = 1 + 3 + 5 + \dots + (2k - 1) + (2(k+1) - 1)$$

Let $k^2 = 1 + 3 + 5 + \dots + (2k - 1)$, then:

$$P(k+1) = \sum_{i=1}^{k+1} 2i - 1 = \boxed{k^2 + (2k + 1)}$$

For the right hand side: $(k+1)^2 = \boxed{(k+1)^2}$

Solution: $P(k+1) = k^2 + (2k + 1) = (k+1)^2$

d.

In a proof by mathematical induction that the formula holds for every integer $n \geq 1$, we suppose that the formula is true for a particular but arbitrarily chosen integer k , and if true, it must also be true for $k + 1$, for both the left-hand side and the right-hand side.

Solution: See above.